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Organic Material: The Primary Control on Mercury Methylation and Ambient Methyl Mercury Concentrations in Estuarine Sediments

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Estuarine environments that have no direct sources of mercury (Hg) pollution may have sediment concentrations of methylmercury (MeHg) as high as those of polluted marine environments. In this study we examined the biogeochemical factors affecting net methylation and sediment MeHg concentrations in an unpolluted estuarine environment, the Öre River estuary, which discharges into the Bothnian Bay (20−120 ng total Hg g^{-1} dry sediment, salinity 3−5‰). We analyzed the spatial and temporal differences in surface sediment profiles of MeHg concentration, Hg methylation, MeHg demethylation, and concentrations of sulfide and oxygen between accumulation and erosion type bottoms. The main difference between the bottoms studied was in the proportion of organic material (OM) in the sediment, ranging between 0.8% and 10.8%. The pore water sulfide concentration profiles also differed considerably between sites and seasons, from 0 to 20 μM, with 100 μM as the extreme maximum. The sediment MeHg concentration profiles (0−10 cm) mostly varied between 0.1 and 7 ng g^{-1} dry weight (dw, as Hg). The MeHg demethylation rates were relatively low and the depth profiles of the rates were relatively constant over season, site, and depth. In contrast, both rates and depths of maximum Hg methylation differed between the bottoms. The results indicate that the amount of OM accumulated at the bottoms was the main factor affecting net MeHg production, while the total amount of Hg had little or no influence on the amount of MeHg in the sediment.

Introduction

Methylmercury (MeHg) is one of the most toxic mercury (Hg) species (*1*). Its abundance in pristine aquatic systems is controlled by two counteracting microbiological processes: mercury methylation (*2*) and methylmercury demethylation (*3*). In aquatic sediments, mercury methylation activity is restricted mainly to sulfate-reducing bacteria (SRB) (*4*−*9*). In contrast demethylation of MeHg is conducted by a range of different bacterial strains in sediments either through reductive demethylation, by expression of the mercury resistance operon (*mer*) (*10*), or through the less-characterized oxidative demethylation mechanism (*11*, *12*).

Hg (Hg^{2+}) availability is a fundamental prerequisite for Hg methylation (*13*). Despite this the correlation between Hg^{2+} and MeHg is weak (*11*, *14*−*18*) and factors affecting SRB activities are as important. The availability of sulfate (electron acceptor) and high quality carbon (electron donor) are the two major variables affecting SRB populations and activities. Hg methylation rates in terrestrial environments are therefore normally limited by sulfate availability (*19*). In marine environments the pool of available sulfate is practically unlimited. This, however, commonly results in high sediment sulfide concentrations, which severely limit Hg methylation (*20, 21*). High (>5−10 μM) sulfide concentrations decrease the rate of methylation, probably through the formation of charged HgS complexes (*22*, *23*). However, low ($\approx\mu$M) levels of sulfide in sediments promote Hg methylation through the formation of neutral Hg−S complexes which is the form that penetrates bacterial (SRB) cell membranes diffusively rather than by way of a dedicated Hg^{2+} cellular transport mechanism (*22*). Estuaries and brackish water environments are characterized by low but nonlimiting access to sulfate. Together with sulfate fertilized terrestrial wetlands (*19*, *24*), estuaries and brackish water environments therefore constitute possible "hotspot" environments for Hg methylation.

At optimal conditions, with respect to sulfate, the availability of fresh organic substrate should theoretically be a dominant factor affecting Hg methylation by SRB in brackish sediments. The bottom topography governs much of the distribution of sediment organic material (OM), which must be sufficiently high for redox conditions allowing sulfate reduction to develop, and the seasonality in primary production is the main factor affecting the temporal addition of fresh OM to the sediment surfaces.

Recent findings indicate that the total Hg (Hg_{tot}) concentration and redox potential influence the relative importance of reductive and oxidative demethylation pathways. While reductive demethylation seems to predominate in Hg contaminated waters under aerobic conditions (*25*, *26*) oxidative demethylation occurs mainly in unpolluted anoxic sediments (*12*, *27*). Oxidative demethylation is presumed to result in Hg^{2+} as an end product, so no net elimination of Hg^{2+} takes place in this process, in contrast to reductive demethylation. Pristine anoxic sediments may therefore be subjected to higher degrees of MeHg accumulation compared to Hg contaminated sediments as a result of substrate (Hg^{2+}) recycling.

In this study, we examined an unpolluted estuarine environment to elucidate the main biogeochemical factors affecting sediment MeHg concentrations, by monitoring the temporal and spatial variations in OM and sulfide contents, as well as Hg methylation/MeHg demethylation rates and correlated these variables to ambient MeHg depth profiles. Five sites with differing bottom characteristics (accumulation and erosion bottoms) were sampled on three occasions during the course of a year to ensure that the samples covered large natural variations in OM and sulfide contents.

Experimental Section

Field Site. The study was conducted in the Öre River estuary (63°30′45″ N, 19°47′81″ E), which is situated in the Norrbyn

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archipelago and discharges into the Bothnian Bay (see ref *28* for graphic details of the field site). The estuary is sheltered and shallow (mean depth between 10 and 20 m) with clay sediments over substantial parts of its bottoms, which are unaffected by any local anthropogenic Hg pollution. Hg_{tot} concentrations in the sediment vary between 20 and 120 ng g^{-1} dry sediment (*28*, *29*). Ice covers the area from mid-December to early May and the salinity is normally 3−5‰. Five sampling sites were selected: designated Accumulation Bottom 1 (AB1), Accumulation Bottom 2 (AB2), Erosion Bottom (EB), Shallow Sandy Bay (SL), and Deep-Hole Accumulation Bottom (DH).

Sampling. Sediment cores were collected with a twin-barrel core sampler (*29*) deployed from the research vessel M/S *Lotty* in October 2001, May 2002, and June 2002. The core liners were immediately capped at both ends with rubber stoppers upon surfacing, leaving approximately 20 cm of seawater on top of the sediments. An extruder/core slicer unit was used for subsampling the sediment cores.

Pore Water Sulfide and Oxygen Saturation Measurements. Pore water sulfide and oxygen saturation depth profiles were determined in the sediment cores within 4 h of arriving at the laboratory using microelectrodes (*29*).

Organic Material, Total Carbon, and Nitrogen Measurements. The OM content of all samples was determined by measuring the gravimetric loss on ignition (LOI) at 550 °C over 4 h (*30*). The precision of LOI measurements was 4% RSD ($n = 10$). Total carbon and nitrogen contents of the samples were measured using a Carlo Erba 1108 elemental analyzer.

Hg_{tot} Measurements. Hg_{tot} was measured using a Leco AMA 254 mercury analyzer fitted with a 44-position automatic solid sampler. The accuracy of measurements was continuously checked by analyzing marine sediment reference material MESS-2 (National Research Council of Canada) at random positions in the sample queue (determined value, 96 ± 5 ng g^{-1} dw (mean ± 1SD, $n = 10$); certified value, 92 ± 9 ng g^{-1} dw).

Hg Methylation, MeHg Demethylation, and Ambient MeHg Determination. All MeHg related concentrations and transformation rates are presented in terms of Hg on a dry weight basis. Hg methylation/demethylation rates and ambient MeHg concentrations were measured using isotope-enriched tracers of Hg^{2+} and $MeHg^{+}$ combined with species-specific isotope dilution mass spectrometry according to a previously described protocol with known accuracy and precision (*29*). Briefly, the top 10 cm of the sediment cores were cut into 1-cm segments, which were transferred to 125 mL Teflon bottles and mixed immediately with 50 mL of degassed seawater from the respective sampling location. The sediment slurries were spiked with aqueous solutions of $Me^{198}Hg^{+}$ and $^{201}Hg^{2+}$ to obtain estimated sediment dry weight concentrations of 1 and 20 ng g^{-1}, respectively. Depending on the sampling location and actual dry weight of the individual sediment sections the spiked tracer concentrations corresponded to 50−100% of ambient Hg species concentrations. Due to the differences in actual dry weight of individual segments some variation in the added tracer concentrations among samples was unavoidable. The slurries were then incubated under N_2 in an anaerobic glovebox at 4 °C in the dark for 10 days. Incubation was stopped by freezing at −20 °C, and followed by lyophilization, sample preparation, and analysis. In a preceding time-cause study using surface sediment from the DH site and a 48 h sampling interval, demethylation and methylation of added tracers was found to progress linearly at about 0.1 ng g^{-1} day^{-1} dw ($r^2 = 0.90$, $n = 6$) and 0.5 ng g^{-1} day^{-1} dw ($r^2 = 0.93$, $n = 6$), respectively, for at least 10 days under these conditions. To obtain comparable results among depths, sites, and occasions methylated/demethylated amounts were normalized with respect to samples containing the highest spiking concentrations of the respective tracers within the sample batch. The method precision for ambient MeHg and methylation/demethylation determinations was within 2−3% RSD, based on replicated subsample analyses ($n = 15$) and subsample incubations ($n = 9$), respectively. The method detection limit for MeHg measurements was calculated to be 0.02 ng g^{-1}.

The measured methylation and demethylation rates presented in this study should be regarded as methylation/demethylation potentials since the added isotopically labeled tracers ($Me^{198}Hg^{+}$ and $^{201}Hg^{2+}$) probably exhibited higher bioavailability than corresponding ambient species. Hence the measured methylation/demethylation rates may therefore not reflect actual in-situ rates.

Statistical Analysis. Partial least squares modeling (PLS) (*31*) was applied separately for each site and sampling occasion as well as jointly for the total data set to examine whether the variables Hg methylation, MeHg demethylation, OM content, and sulfide concentration correlated with the ambient MeHg depth profiles. For this purpose SIMCA-P+ 10.0 was used (Umetrics AB, Umeå, Sweden).

Results

Organic Material, Total Carbon, and Total Nitrogen Contents. The OM content in the sediments varied by more than a factor of 10 among sampling sites, from around 1% at the SL site to about 11% at the deep-hole accumulation bottom (Table S1, Supporting Information). The variation at individual sites was relatively small among the different sampling occasions, one exception being the EB site in May (3.1 vs 5.2 and 5.6% in October and June, respectively). The AB 1, AB 2, and DH sites displayed similar total carbon (C_{tot}) and nitrogen (N_{tot}) contents, as relative proportions of OM contents, which remained quite constant among the different sampling occasions. Corresponding values for the EB and SL sites were similar, albeit with somewhat more seasonal variation.

Pore Water Sulfide and Oxygen Saturation Concentrations. The shallow sandy bay sediments consisted of quite large-grained sand, so it was not possible to measure oxygen and sulfide profiles in these samples with the available microelectrodes without the risk of breaking them. Oxygen profiles for all other samples on all occasions exhibited practically identical properties. Sediment surface oxygen saturations of about 95% were reduced to 0% descending, at most, 5 mm into the sediments. Sulfide depth distributions were relatively similar at the two accumulation bottoms and the erosion bottom, while the maximum values at the deep-hole accumulation bottom were considerably higher than at any of the other sites. The vertical concentration profiles were low and relatively constant at all sites in May, while a steep increase from the surface downward occurred in October at all sites except for the deep-hole accumulation bottom were the concentration profile remained fairly constant over depth. At this site (DH) sulfide levels peaked in June at close to 100 μM near the surface and again at a depth of approximately 6.5 cm (Figure 1).

MeHg and Hg_{tot} Field Concentrations. Ambient MeHg concentrations varied considerably with bottom type, depth below the sediment surface, and season (Figure 2), with the levels ranging between ~0.1 and 13 ng g^{-1}. The highest levels were in most cases found in October and the lowest were found in May. Compared to the other sites, the SL site displayed quite low and vertically homogeneous concentrations that differed little among sampling occasions. MeHg concentrations at the erosion bottom (EB) increased with depth from approximately 0.3 ng g^{-1} at 0.5 cm to 0.6 ng g^{-1} at 9.5 cm in May. In June the corresponding values were 0.4 ng g^{-1} and 1.6 ng g^{-1}. At the AB 1 site MeHg levels peaked at depths between 2 and 5 cm at 2 ng g^{-1} in May and about

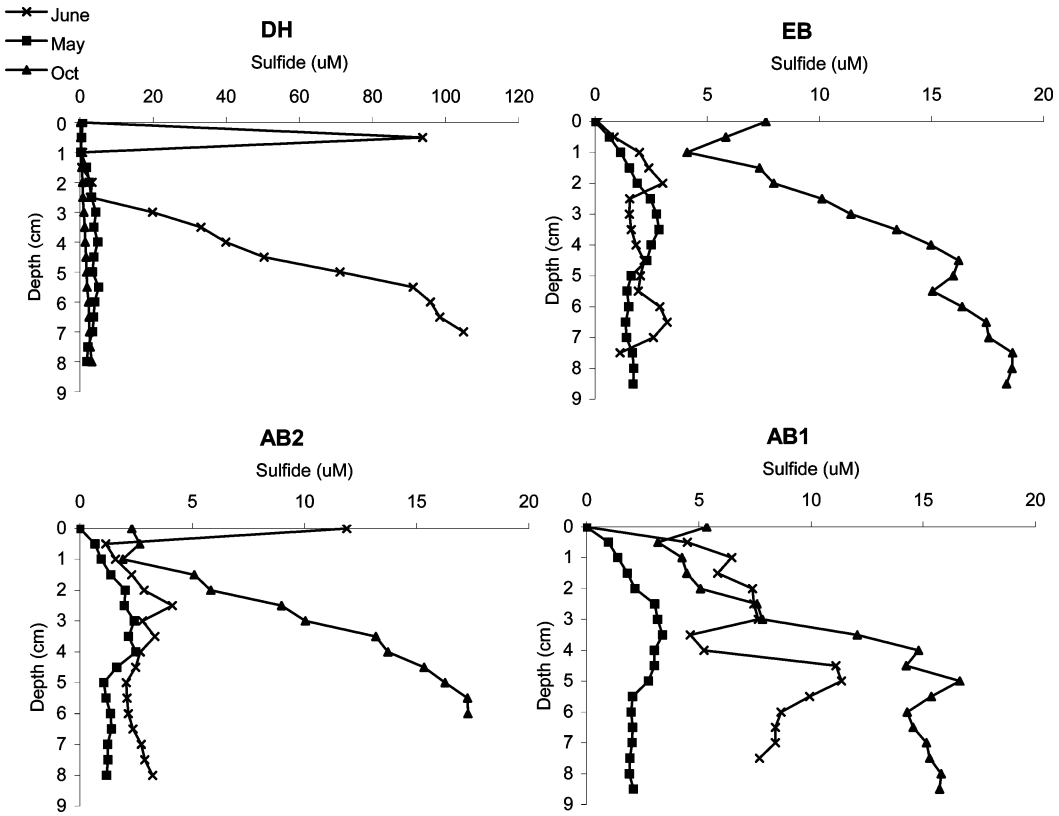


FIGURE 1. Sulfide depth profiles in surface sediments at four of the sampling sites on the three sampling occasions (DH, deep-hole accumulation bottom; EB, erosion bottom; AB (1 and 2), accumulation bottoms).

2.5 and 3.5 ng g^{-1} in June and October, respectively. The vertical distribution of MeHg was slightly different at the AB 2 site, with two regions at 1−4 cm and 7−9 cm reaching roughly 2 ng g^{-1} on all three sampling occasions. The highest ambient levels of MeHg were found at the DH site, for which a concentration of 13.2 ng g^{-1} was recorded in the 0−1 cm surface segment in June.

Ambient Hg_{tot} varied between 20 and 120 ng g^{-1} (dw) with the depth profiles being constant at the SL (20 ng g^{-1}) and EB (40 ng g^{-1}) sites and increasing with depth at AB1 (70−100 ng g^{-1}), AB2 (60−80 ng g^{-1}), and DH (80−115 ng g^{-1}) sites. Only moderate variation occurred among sampling occasions (Figure S1, Supporting Information). Variation in the sediment organic material content explained 67% of the variation in Hg_{tot} (Figure S2, Supporting Information).

Hg Methylation. The highest rates of Hg methylation were observed in June at all sites, except for the SL site, and the lowest rates were observed in May (Figure 2). Each individual sampling site showed a characteristic Hg methylation depth profile that remained fairly constant among sampling occasions. The methylation rates in the shallow sandy bay were low (mostly <0.2 ng g^{-1} day^{-1}) and their depth profiles were quite constant. The methylation rates were maximal in the erosion bottom at 6−8 cm with a methylation rate of 0.2−0.6 ng g^{-1} day^{-1}; in the two accumulation bottoms at 2−4 cm and 0.3−0.8 ng g^{-1} day^{-1}; deep-hole accumulation bottom, 0−1 cm and 0.5−2.3 ng g^{-1} day^{-1}. The highest overall methylation rate was observed at the DH site (2.3 ng g^{-1} day^{-1} in the 0−1 cm segment in June).

MeHg Demethylation. MeHg demethylation depth profiles were relatively consistent among sampling occasions and sites in contrast to the corresponding methylation profiles (Figure 2). Overall, the June and October samples differed marginally in demethylation activity and the rates on these occasions were about a factor of 2 higher than those in the May series. All sampling sites exhibited comparable demethylation rates within sampling occasions and the rates were relatively constant vertically. In some samples the demethylation activity was slightly increased in the top 3−4 cm of the sediment core, for example in June measurements of AB 1, AB 2, and EB as well as the EB site in the October series.

Net Methylation Potential. Net methylation potentials were derived by subtracting the amount of demethylated $Me^{198}Hg$ tracer from the amount of $Me^{201}Hg$ generated from the $^{201}Hg^{2+}$ tracer. This procedure provided an identifying measure on the dominating process (methylation/demethylation) in a specific sample. As a result of the relatively constant demethylation rates the net methylation potential profiles (Figure 2) reflected to a large extent the measured methylation profiles. One exception was the SL site in which demethylation rates exceeded methylation rates, making it a net sink of MeHg. The highest observed net methylation potential was observed in the 0−1 cm segment at the DH site in June (2.2 ng g^{-1} day^{-1}).

Factors Affecting Sediment MeHg Depth Profiles. The vertical distribution of MeHg at each site and sampling occasion was significantly ($p < 0.05$) explained by different combinations of the variables organic material content (OM), sulfide concentration (S^{2-}), Hg methylation rate (Meth.), and MeHg demethylation rate (Demeth.) in partial least-squares (PLS) regression models (Table 1) while Hg_{tot} was not significant and therefore not included in the model. The fitted models generally accounted for more than 85% of the variance in MeHg, while the variance explained by the cross-validated models was less consistent. The three models that explained the least variance, among both the best fitted and the cross-validated models, were all for the May sampling occasion. For 5 out of 6 models with significant ($p < 0.05$) regression coefficients for methylation, the coefficients were positively related to the MeHg concentration. In addition, 4 out of 5 significant coefficients for demethylation were positively related to the MeHg concentration. The sulfide concentration

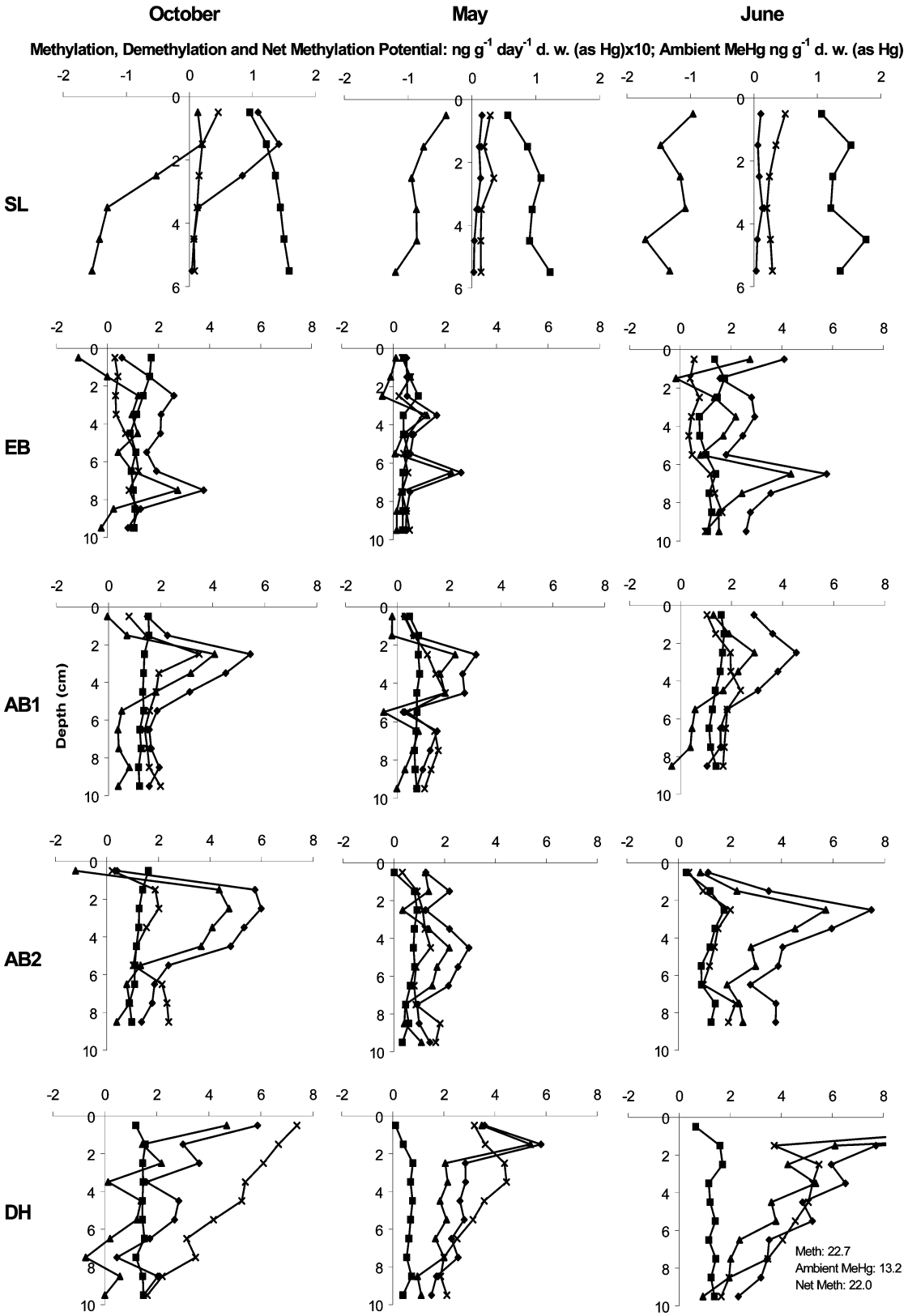


FIGURE 2. Depth profiles of ambient MeHg concentration (crosses), Hg methylation (diamonds), MeHg demethylation (squares), and net methylation (triangles) from the five sites and three measurement occasions (site codes as in Figure 1 with, also, SB, shallow sandy bay).

contributed significantly ($p < 0.05$) in 5 of the models, 4 of which were derived from data collected from the accumulation bottoms with regression coefficients that were all positively related to MeHg. The fifth model was derived from samples from the deep-hole accumulation bottom and the significant ($p < 0.05$) regression coefficient was negatively correlated to MeHg. All but one of the significant regression coefficients for OM content were positively correlated to MeHg.

A total data set PLS model based on the same variables explained 69% of the variance in ambient MeHg with Hg methylation and OM content as significant positive coef-

TABLE 1. Proportion of Variance Explained by Best-Fit (R^2) and Internal Cross-Validated (Q^2) PLS Regression Models, and Regression Coefficients, Based on Data from Each Segment, Site, and Sampling Occasion

				PLS regression coefficients[a]			
site	occasions	R^2 [b]	Q^2 [c]	meth.	demeth.	S^{2-}	OM
SL	May	0.86	0.64	**0.65**[d]	−0.50	n.d.[e]	**0.54**
	June	0.81	0.42	**−0.42**	0.08	n.d.	0.97
	October	0.98	0.94	**0.27**	**0.90**	n.d.	−0.39
EB	May	0.70	−0.2	0.52	0.67	0.97	0.78
	June	0.88	0.51	**0.74**	−0.39	−0.03	**−0.44**
	October	0.92	0.68	−0.27	−0.15	1.45	0.63
AB1	May	0.52	0.34	**0.32**	−0.14	**0.28**	−0.15
	June	0.99	0.97	0.07	**0.79**	**0.64**	**0.73**
	October	0.96	0.36	0.89	0.87	−0.30	0.54
AB2	May	0.30	−0.10	0.03	−0.28	**0.29**	0.00
	June	0.92	0.70	**0.25**	**−0.43**	**0.34**	−0.43
	October	0.96	0.83	0.45	**2.38**	−1.09	1.08
DH	May	0.86	0.80	0.49	−0.16	−0.21	**0.92**
	June	0.89	0.61	**0.41**	**0.33**	−0.09	**0.30**
	October	0.98	0.85	−0.02	0.15	**−1.01**	0.04
all	all	0.69	0.57	**0.53**	0.07	0.09	**0.46**

[a] Meth. = methylation rate; Demeth. = demethylation rate; S^{2-} = sulfide concentration; OM = organic material content. [b] Explained variance based on the best estimated model. [c] Model predictivity based on explained variance from internal cross-validation. [d] Bold-face regression coefficients represent significance at $p < 0.05$. [e] Not determined.

ficients. The cross-validated model accounted for 57% of ambient MeHg variance. A linear regression of MeHg on the total mercury content (Hg_{tot}) gave an r^2 value of 0.28. After removing all co-variation between organic material and Hg_{tot} and between organic material and MeHg, respectively, a regression of residual ambient MeHg on residual Hg_{tot} gave no explained variance.

Discussion

Hg_{tot} and MeHg Levels. Background levels of Hg_{tot} in sediments are governed by the amount deposited, enrichment processes, and release to the overlying water column. Even Hg background levels should therefore be assumed to vary and background levels are commonly regarded as being <100 ng g^{-1} dry sediment. Estimates of pre-industrial sediment levels include the following: <20 ng g^{-1} (dw) in a semi-enclosed macro-tidal estuary in southeastern Canada (*16*); 20 ng g^{-1} (dw) in the Bothnian Bay; 30 ng g^{-1} (dw) in the Bothnian Sea (*32*); and 50−100 ng g^{-1} (dw) in an open estuarine environment south of Spain (*33*). The Hg_{tot} concentrations in the brackish Öre River estuary were 20−120 ng Hg g^{-1} (dw), with the highest values occurring in the accumulation and deep-hole accumulation bottom sites (*28*, *29*), and can therefore be regarded as background levels. Despite the low Hg_{tot} concentrations, a large proportion of the sediment MeHg concentrations seem to be relatively high compared to those found in other marine environments that have not been contaminated by Hg from any anthropogenic point source (*16*, *28*, *34* and Table S4, Supporting Information). The ambient MeHg concentrations found in this study are actually similar to or even higher than those normally encountered in Hg-contaminated sediments (*11*, *17*, *18*, *35*−*37*). The MeHg/Hg_{tot} ratios, based on Hg_{tot} data presented in Figure S1, Supporting Information, were in many cases unexpectedly high (∼4−13%).

Since Hg is the substrate for Hg methylation, Hg_{tot} in sediments is expected to correlate with MeHg (*13*, *14*). However, based on compiled sediment data on Hg_{tot} and MeHg from a wide range of aquatic environments the relationship was found to be quite weak ($r^2 = 0.40$) with a 95% prediction confidence interval spanning 2 orders of magnitude (*13*). Sediment data from San Francisco Bay (*17*), covering both contaminated and uncontaminated sites ($n = 25$), revealed no correlation between Hg_{tot} and MeHg. Low or nonexisting correlation between concentrations of Hg_{tot} and MeHg was also found, both within data in this study ($r^2 = 0.28$, Figure S3, Supporting Information), as well as in other studies (Table S4, Supporting Information), and data compiled by Kannan and Falandysz (*34*). The low r^2 value found in this study was reduced to zero when the effect of co-variation with organic material was removed, further supporting the idea that the total amount of Hg_{tot} in the sediment had very little effect on the MeHg concentrations. These findings further highlight the importance of biogeochemical controls on MeHg concentrations. The main biogeochemical characteristics of the studied area, the Öre River estuary, are that the salinity and sulfate concentrations in the overlying water column are relatively low (3−5‰ and 3 mM, respectively) and the OM content is relatively high (Table S1, Supporting Information).

Sulfate and Sulfide Effects on MeHg and Methylation. The water column chloride and sulfate concentrations in the estuary (∼70 and ∼3 mM, respectively) are close to optimal for microbial Hg methylation (*20*, *38*, *39*). This makes coastal environments and estuaries in the Bothnian Sea important potential sources of MeHg. Low sulfate availability limits the access of the sulfate-reducing bacteria (SRB) to electron acceptors, while high sulfate concentrations provide the potential for high sediment sulfide concentrations limiting Hg methylation (*13*, *20*, *22*). Experimentally increased sulfide concentration to 5 μM decreased methylation by 20% relative to the control (1 μM sulfide), while at 350 μM sulfide in the sediment pore water only 10% of the original methylation activity remained (*13*).

Since the sediment pore water concentration of sulfide in our study was generally between 2 and 20 μM (with levels up to 100 μM as an exception) the rate of Hg methylation was probably only marginally lowered by in-situ sulfide concentrations. In 4 out of 5 significant PLS regression models for specific sites and sampling occasions the correlation between the rate of methylation and sulfide levels in the vertical profiles was positive. These results from the Öre River estuary sediments is consistent with the hypothesis that methylation is facilitated by the presence of neutral Hg−sulfide complexes (*22*, *23*) and is not counteracted by the formation of charged HgS complexes (*22*) as is commonly the case at the sulfide concentrations occurring in open marine environments (*38*).

Demethylation. The low and relatively constant rates of demethylation found in this study (0.05−0.2 ng g^{-1} day^{-1}) are in contrast to the much higher and more variable levels found in various other environments, e.g., Hg-polluted sediments in the Gulf of Trieste (*11*). Our results suggest that neither OM nor sulfide contents in brackish water sediments exert any significant influence on MeHg demethylation rates. Of the two biotic pathways identified for MeHg demethylation the oxidative demethylation pathway does not appear to be an active detoxification mechanism (*13*), indicating that the rate of demethylation is unlikely to be influenced by the concentration of MeHg, as in the case of reductive demethylation (*26*). Oxidative demethylation also occurs in a number of both aerobic and anaerobic microorganisms (*12*, *40*, *41*), suggesting that potential rates of demethylation are unlikely to be highly dependent on microbial community composition. This is consistent with the relatively even vertical distributions of demethylation rates we found, despite the pronounced zonation of microbial communities in the sediments. A third demethylation mechanism was recently suggested by Martin-Doimeadios et al. (*15*), who demon-

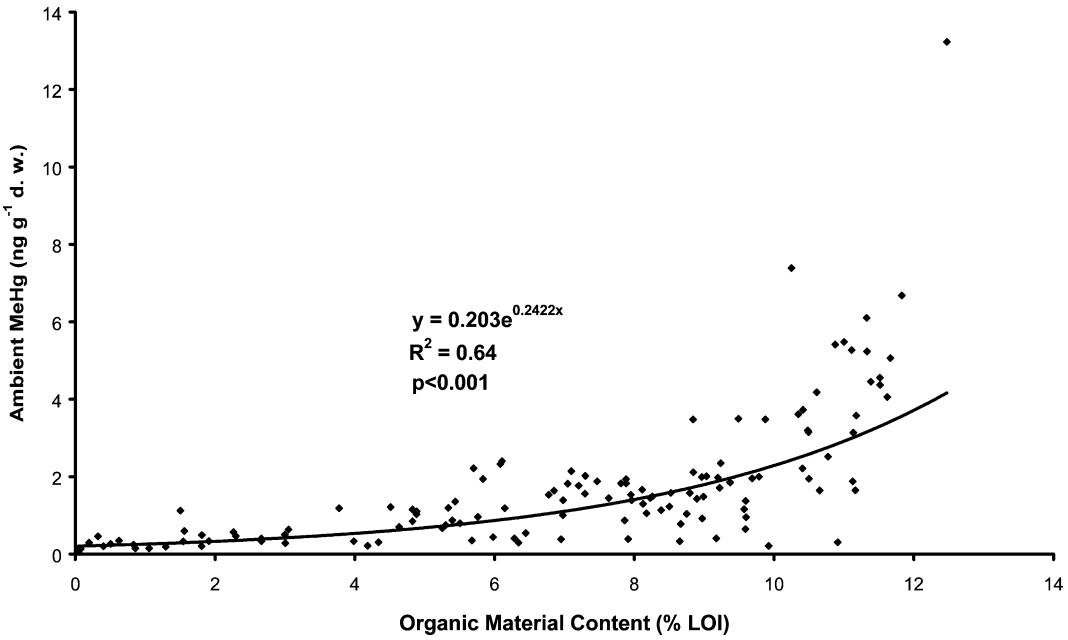


FIGURE 3. Ambient MeHg concentrations plotted against the organic material content for all samples in the study; 64% of the variation in MeHg concentrations was explained by the percentage of organic material according to an exponential function.

strated that MeHg demethylation rates after autoclaving (121 °C, 20 min) estuarine sediments were as high as those found under biotic conditions, suggesting that abiotic processes might be important for demethylation. Even if abiotic demethylation is most likely affected by the physicochemical properties of the sediment, abiotic demethylation is probably less variable than biotic demethylation.

Influence of Organic Material on Hg Methylation. Provided that the availability of sulfate is within the optimal range (*20*), and both Hg_{tot} concentrations and rates of demethylation are low (and temporally and spatially relatively constant), as in the estuary studied, then the main factors accounting for variations in both ambient MeHg levels and rates of methylation should theoretically be differences in bottom characteristics and the supply of OM to the SRB community. The proportion of OM in the sediment (ranging from 0.04 to 13%) was the best single predictor, explaining 64% of the variation in the exponentially transformed MeHg concentrations in the sediments (Figure 3). At an OM content between 8 and 10% the average (±SD) MeHg concentration was 1.6 ± 0.8 ng g^{-1} and at OM contents >10% the corresponding values were 4.2 ± 2.5 ng g^{-1}. The median OM sediment content in the Bothnian Sea and Bothnian Bay is 10% (*32*), indicating that MeHg concentrations could be high (cf. Table S4, Supporting Information) in this brackish water sea. Including data on methylation/demethylation rates and pore water sulfide concentration together with the content of OM in the sediment slightly increased (to 68%) the amount of variation in MeHg concentrations explained by the PLS model (Table 1).

With respect to methylation, the most striking difference between the sampled bottom types is in their vertical distribution pattern of methylation rates, although the rates also differ to some extent (Figure 2). The major differences between the bottom sites are that both water depth and the content of OM in the sediment increases from the shallow sandy bay to the deep-hole accumulation bottom.

We suggest that the main factors influencing the spatial variation in both the rate and depth of maximal methylation in the brackish sediments are the OM content and the supply of alternative electron acceptors, i.e., oxygen from the hypolimnion, while the sedimentation rates of dead bacteri- and phytoplankton, i.e., "high-quality" carbon, are the main influences on the temporal variation. Organic material may affect methylation and MeHg concentrations through several mechanisms. High contents of OM in the sediment are a prerequisite for maintaining low redox potentials while supplies of "high-quality" OM provide electron donors for the SRB. OM also provides complexing agents for Hg^{2+} and MeHg, thus influencing both the total sediment concentrations and the partitioning between solid and dissolved phases. Hammerschmidt et al. (*14*) recently demonstrated increased retention of MeHg to the solid phase with increasing OM content in Long Island Sound sediments.

A high content of OM in the sediment promotes heterotrophic microbial activity, which consumes oxygen and lowers the redox potential close to the sediment surface, cf. the accumulation bottom data. The lower the organic content in the sediment the deeper can oxygen and other competing electron acceptors (mainly manganese and iron) penetrate before they are depleted by heterotrophic microbial activity, cf. the shallow bay and the erosion bottom data. After the potential for heterotrophic microbial consumption of electron acceptors in the sediment, the second most important influence is the supply of oxygen from the epilimnion. The deeper and more sheltered the basins are, the more limited will the supply of oxygen to the sediment be, allowing microbial populations that utilize alternative electron acceptors such as sulfate to thrive closer to the sediment surface. The most important consequences of these variations in depth profiles are their effects on the proportion of organic carbon deposited through sedimentation each year that will be terminally mineralized by sulfate-reducing bacteria. The proportion of the annual input of newly dead phyto- and bacteriplankton that will be terminally mineralized by a SRB population is positively correlated with its closeness to the sediment surface. Therefore, a sediment profile in which the methylation potential is maximal at the sediment surface indicates that a large proportion of the annual accumulation of organic carbon is terminally mineralized by sulfate-reducing bacteria. The exponential relationship between OM content and MeHg concentrations also suggests that OM content tends to enhance MeHg concentrations by two or more different processes that positively interact.

While we suggest that the OM content of the sediment is the main factor influencing the spatial variation in meth-

ylation and MeHg concentrations, we propose that the temporal variations in Hg methylation rates and ambient MeHg concentrations are dependent on seasonal variations in the production of fresh autochthonous carbon (cf. Table S3, Supporting Information), i.e., dead bacteri- and phytoplankton. This hypothesis is supported by the PLS model based on the entire data set with OM and rate of methylation as predictors (Table 1). Methylation rates accounted for additional variation that OM alone could not explain. The seasonal input of fresh organic carbon to the sediment surface is minor in comparison to the amount of OM in the sediment. Temporal differences in OM are therefore not seen in the data (Table 1), but the seasonality in organic carbon sedimentation is the main driver of microbial carbon mineralization at the sediment surface, since it affects rates of processes such as sulfate reduction and thus Hg methylation. Seasonality in the ambient MeHg concentrations and methylation rates can be mainly ascribed to differences in the supply of fresh OM serving as a substrate for the SRB, the transport of oxygen to the bottom sediment through the water column, and the sediment temperature through the activity of the microbial community increasing the consumption rate of alternative electron acceptors.

In summary, it can be concluded that the net production of bio-available Hg, i.e., MeHg, at sites that are not affected by Hg pollution can be as high as that in Hg-polluted areas. This is principally governed by the supply of new OM to the sediment and the redox conditions therein, and these geochemical factors overrule the effect of the total Hg content. A high organic content in the sediment maintains a low redox potential; a prerequisite for sulfate reduction and concomitant Hg methylation. In sediments with lower organic contents Hg methylation occurs deeper in the sediment and requires more stagnant overlying water, i.e., deep water or areas with restricted water circulation. In sediments with high organic content the redox potential required for sulfate reduction can be maintained closer to the sediment surface and the supply rate of alternative electron acceptors from the overlying water may be relatively high. In contrast, this study shows that MeHg demethylation is not significantly affected by either sediment OM content or redox conditions, indicating that this process may, at least partly, be abiotcally driven by factors other than those studied here. In extension, the results presented in this study suggest that estuaries and brackish aquatic environments, with low salinity, constitute environments with a potential risk for high net production of bio-available MeHg, which may be further strengthened if such areas become eutrophic, e.g., via increased anthropogenic nutrient loads.

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Supporting Information Available

Organic material, total carbon, and total nitrogen concentrations in the sediments are given in Table S1. Tables S2 and S3 show the seasonal water column total organic carbon concentration and primary production in the Öre River estuary at the different sampling occasions. Table S4 shows data on Hg_{tot}, MeHg, Hg-methylation, salinity, organic material, and organic carbon (C_{org}) in aquatic sediments compiled from both Hg-polluted and unpolluted estuaries. Figure S1 shows Hg_{tot} depth profiles for the different sampling sites and occasions. Figures S2 and S3 show the correlation between OM:Hg_{tot} and Hg_{tot}:MeHg, respectively, for all samples in the study. This material is available free of charge via the Internet at http://pubs.acs.org.

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